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# Highly Homogeneous Stereocontrolled Construction of Quaternary Hydroxyesters by Addition of Dimethylzinc to α-Ketoesters Promoted by Chiral Perhydrobenzoxazines and $B(OEt)_3$

**Rebeca Infante, Javier Nieto,* and Celia Andrés*[a]**

**Abstract:** A highly efficient enantioselective addition of $Me_2Zn$ to α-ketoesters, assisted by a chiral perhydro-1,3-benzoxazine ligand, is described. This novel catalytic system offers homogeneous elevated enantioselectivities in the preparation of α-hydroxyesters that bear a quaternary stereocenter, with a minor dependence on electronic and steric effects when aromatic, heteroaromatic, or aliphatic α-ketoesters are employed. The catalyst can be recovered and reused without loss of activity.

**Keywords:** alcohols · asymmetric catalysis · ketoesters · quaternary stereocenter · zinc

## Introduction

The asymmetric construction of quaternary stereocenters through the creation of carbon–carbon bonds represents one of the most important tasks in asymmetric synthesis. Especially interesting is the enantioselective formation of chiral tertiary alcohols, which constitute essential "building blocks" for the synthesis of a wide range of enantiomerically pure natural and pharmaceutical products.[1] Catalytic enantioselective addition of a nucleophilic organometallic reagent to a ketone is probably the most efficient method known for this purpose. Nevertheless, although an extensive collection of highly efficient catalysts have been developed to promote the addition of organozinc reagents to aldehydes,[2] comparatively few reports involve ketones as electrophiles.[3,4] For this reason, competent enantioselective additions to ketones still represent a challenge. This might be attributed to their inherently lower reactivity towards these organometallic species and also to difficulties in stereodifferentiation of the two enantiotopic faces of the ketone.

The use of more reactive ketones, such as α-ketoesters, could be an advantage in terms of reactivity, although not for the enantiocontrol. This particular kind of electrophile may act as a chelating ligand and enhance the reactivity of the nucleophilic species, which would favor the uncatalyzed reaction pathway and have a detrimental effect on the enantioselectivity. Notably, the alkylation products of α-ketoesters bear both a quaternary stereocenter and an ester group, hence they are suitable precursors for the synthesis of many valuable complex compounds.[5] Although the use of chiral auxiliaries to promote diastereoselective additions of organometallic reagents to enantiopure α-ketoesters has been widely studied,[6] scarce examples of catalytic asymmetric additions with a chiral ligand have been reported.[7] The first catalyst used to this end was Kozlowski's titanium–salen complex,[7a] which affords the $Et_2Zn$ addition products in good yields, although with moderate enantioselectivities (up to 78% enantiomeric excess (*ee*)). Since then, approximately five different kinds of ligand have been employed for catalytic alkylation of α-ketoesters,[7b–f] two of which have been reported to achieve high enantioselectivities (>90% *ee*). Shibasaki et al. reported a privileged prolinol-derived ligand, which promoted addition of $Me_2Zn$ to α-ketoesters with enantioselectivities ranging from 59–96% *ee*.[7b] In 2005, Hoveyda and Snapper disclosed an aluminum-catalyzed enantioselective addition of $Et_2Zn$ and $Me_2Zn$ to α-ketoesters that permitted the isolation of products with notable enantioselectivity (56–95% *ee*).[7c]

Recently, we have shown that chiral decahydro-4a*H*-pyrrolo[2,1-b,1,3]benzoxazin-3-ol ligands promote asymmetric addition of diethylzinc and arylzinc derivatives to aldehydes with excellent enantioselectivities under mild conditions, even at very low ligand loadings (0.2 mol%).[8] Based on these results, herein we describe the asymmetric addition of dimethylzinc to aromatic, heteroaromatic, and aliphatic α-ketoesters promoted by enantiopure perhydro-1,3-benzoxazine-based ligands. Additionally, we further exploited the versatility of the resulting hydroxyesters for the synthesis of complex valuable products, such as oxazolidinediones or azidoalcohols, which are important building blocks for biologically active compounds.[9]

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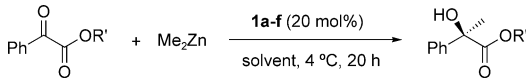
## Results and Discussion

Initially, asymmetric methylation of alkyl benzoylformates **2a–c** by addition of dimethylzinc (6 equiv) in the presence of perhydrobenzoxazines **1a–f** (20 mol%)[8a,10] at 4°C was chosen as a reaction model to evaluate both the catalytic activity of these ligands and the optimal alkoxy substituent (OR′) of the ester. The results are collected in Table 1.

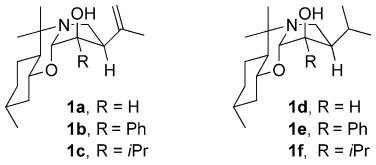
Table 1. Asymmetric methylation of benzoylformates **2a–c**: screening of perhydrobenzoxazine ligands **1a–f** and the alkoxy substituent of the ester.




**2a-c** **3a-c**

| Entry | 2 | R′ | Ligand | Solvent | 3 | Yield [%][a] | *ee* [%][b] |
|---|---|---|---|---|---|---|---|
| 1 | **2a** | Me | **1a** | toluene | **3a** | 95 | 0 |
| 2 | **2a** | Me | **1b** | toluene | **3a** | 90 | 24 |
| 3 | **2a** | Me | **1c** | toluene | **3a** | 93 | 53 |
| 4 | **2a** | Me | **1d** | toluene | **3a** | 91 | 12 |
| 5 | **2a** | Me | **1e** | toluene | **3a** | 78 | 30 |
| 6 | **2a** | Me | **1f** | toluene | **3a** | 55 | 25 |
| 7 | **2b** | Et | **1c** | toluene | **3b** | 98 | 62 |
| 8 | **2c** | *i*Pr | **1c** | toluene | **3c** | 94 | 37 |
| 9 | **2b** | Et | **1c** | hexane | **3b** | 91 | 74 |

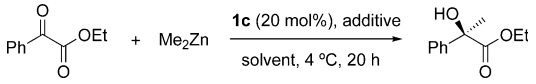
[a] Yield of isolated product. [b] Determined by chiral HPLC.




All of the ligands tested led to hydroxyester **3a** in good yields, although poor to moderate enantioselectivities were found (Table 1, entries 1–6). The influence of the R group of the benzoxazine (**1a–c**) revealed that the stereocenter substitution is critical to obtain good selectivities; a clear improvement of enantiocontrol was observed by increasing the steric volume (Table 1, entries 1–3). Unexpectedly, an increase of the steric hindrance in the vicinity of the hydroxyl group (isopropenyl to isopropyl group; ligands **1d–f**) did not improve the enantioselectivity of the methylation process (Table 1, entries 4–6). To our surprise, a modest *ee* was achieved with the aminoalcohol **1e** (Table 1, entry 5), which has proven to be an excellent chiral ligand for the addition of diethylzinc and arylzinc derivatives to aldehydes.[8] The effect of the alkoxy substituent on the ester was examined in the presence of ligand **1c**, under the same reaction conditions in toluene and hexane (Table 1, entries 7–9). Whereas the bulky ketoester **2c** yielded the corresponding product **3c** with poor selectivity (Table 1, entry 8), a significant increase in asymmetric induction was found when ethyl ester **2b** was used, especially when the reaction was performed in hexane (Table 1, entries 7 and 9). The hydroxyester **3b** could be obtained in excellent yield (91%) and moderate enantioselectivity (74% *ee*).

To enhance the enantioselectivity of the process, we attempted to exploit the effect of a set of additives on the methyl group transfer to ethyl benzoylformate.[11] The experiments were carried out with different amounts of commercially available achiral additives in the presence of dimethylzinc (6 equiv) and **1c** (20 mol%) at 4°C for 20 h (Table 2). Analysis of the results revealed that all screened

Table 2. Effect of additives on asymmetric addition of $Me_2Zn$ to ethyl benzoylformate (**2b**).




**2b** **3b**

| Entry | Additive | [mol%] | Solvent | Yield [%][a] | *ee* [%][b] |
|---|---|---|---|---|---|
| 1 | DiMPEG | 10 | toluene | 77[c] | 81 |
| 2 | *i*PrOH | 25 | toluene | 90 | 82 |
| 3 | $Ti(OiPr)_4$ | 25 | toluene | 73[d] | 75 |
| 4 | $AlMe_3$ | 25 | toluene | 93 | 80 |
| 5 | $BEt_3$ | 25 | toluene | 75 | 65 |
| 6 | $B(OEt)_3$ | 25 | toluene | 94 | 84 |
| 7 | $B(OEt)_3$ | 25 | hexane | 96 | 73 |

[a] Yield of isolated product. [b] Determined by chiral HPLC. [c] 20% of starting material was recovered. [d] 22% of isopropyl hydroxyester **3c** was isolated due to transesterification under the reaction conditions.

additives (either Lewis or Brønsted acids) afforded the target product **3b** in good yield and with improved enantioselectivity. The best result was achieved in toluene when triethyl borate (25 mol%) was used (84% *ee*; Table 2, entry 6). This additive generates trimethyl borane in the presence of dimethylzinc, to deliver zinc ethoxide to the reaction medium,[12] which could increase the reactivity of the nucleophile in the catalytic intermediate, thus beneficially reducing the effect of the background reaction.

Further investigations into several reaction parameters (solvent, temperature, slow addition of the ketoester, catalyst loading, amount of $B(OEt)_3$) were carried out (the results are summarized in Table 3). With ligand **1c**, hexane did not offer any benefit relative to toluene at −10°C (Table 3, entry 1 vs. 2), therefore we continued our study with toluene as a solvent. Temperature variations (Table 3, entries 1, 3, 4, and 7) indicated a maximum enantioselectivity for the reactions performed at −35°C; product **3b** was obtained almost quantitatively with excellent *ee* (Table 3, entry 4). In this context, when the reaction was run at lower temperature (Table 3, entry 7) no beneficial effects were observed on the enantiocontrol and the reaction time was extended to 48 h to obtain the product in good yield. The slow addition of the α-ketoester over 20 h did not improve the enantioselectivity either (Table 3, entries 6 and 7). A considerable decrease in *ee* was perceived when the catalyst loading was reduced to 10 mol% (Table 3, entry 9), although the use of

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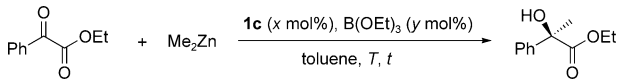
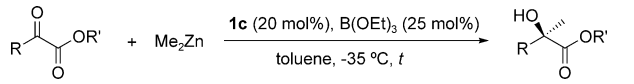
Table 3. Optimization of the reaction conditions.

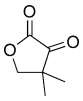
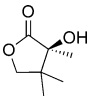



| Entry | **1c** [mol%] | $B(OEt)_3$ [mol%] | *T* [°C] | *t* [h] | Yield [%][a] | *ee* [%][b] |
|---|---|---|---|---|---|---|
| 1 | 20 | 25 | −10 | 20 | 98 | 89 |
| 2[c] | 20 | 25 | −10 | 20 | 97 | 89 |
| 3 | 20 | 25 | −20 | 20 | 98 | 91 |
| 4 | 20 | 25 | −35 | 20 | 97 | 94 |
| 5[d] | 20 | 25 | −35 | 20 | 97 | 95 |
| 6[e] | 20 | 25 | −35 | 24 | 93 | 95 |
| 7[f] | 20 | 25 | −50 | 48 | 89 | 93 |
| 8 | 10 | 25 | −35 | 40 | 66 | 73 |
| 9 | 10 | 12.5 | −35 | 50 | 60 | 85 |
| 10 | 15 | 18.7 | −35 | 40 | 95 | 90 |

[a] Yield of isolated product. [b] Determined by chiral HPLC. [c] Hexane was used as the solvent. [d] The reaction was carried out by addition of **2b** over 30 min. [e] The reaction was carried out by slow addition of **2b** over 20 h and was stirred for a further 4 h. [f] The reaction was performed by slow addition of **2b** over 20 h and was stirred for a further 28 h.

Table 4. Scope of the asymmetric addition of $Me_2Zn$ to α-ketoesters **2**.




| Entry | **2** | R′ | R | *t* [h] | **3** | Yield [%][a] | *ee* [%][b] |
|---|---|---|---|---|---|---|---|
| 1 | **2a** | Me | Ph | 18 | **3a** | 94[c] | 93 |
| 2 | **2b** | Et | Ph | 20 | **3b** | 97 | 95 |
| 3 | **2c** | *i*Pr | Ph | 21 | **3c** | 95 | 95 |
| 4 | **2d** | Et | *o*-Me-Ph | 96 | **3d** | 36[d] | 85 |
| 5 | **2e** | Et | *o*-MeO-Ph | 90 | **3e** | 48[e] | 96 |
| 6 | **2f** | Et | *p*-Me-Ph | 24 | **3f** | 82 | 91 |
| 7 | **2g** | Et | *p*-MeO-Ph | 72 | **3g** | 91 | 94 |
| 8 | **2h** | Et | *p*-Cl-Ph | 20 | **3h** | 71 | 93 |
| 9 | **2i** | Et | 3,4-$(MeO)_2$-Ph | 90 | **3i** | 67[f] | 90 |
| 10 | **2j** | Et | 2-naphthyl | 22 | **3j** | 93 | 91 |
| 11 | **2k** | Et | 2-thienyl | 24 | **3k** | 80 | 90 |
| 12 | **2l** | Et | $PhCH_2CH_2$ | 24 | **3l** | 96 | 84 |
| 13 | **2m** |  | | 40 | **3m**  | 75 | 86 |

[a] Yield of isolated product. [b] Determined by chiral HPLC. [c] Transesterification occurred under the reaction conditions to give a mixture of **3a** and **3b** (82:18) in 93% *ee* for both compounds. [d] 60% of starting material was recovered. [e] 45% of starting material was recovered. [f] 28% of starting material was recovered.

15 mol% resulted in a very competitive 90% *ee* (Table 3, entry 10). In addition, the catalytic amount of triethyl borate seems critical, not only in terms of enantioselectivity, but also for the chemical yields (Table 3, entries 8 and 9). In summary, the optimal conditions found for the methylation of ethyl benzoylformate with $Me_2Zn$ (**1c** (20 mol%) and triethyl borate (25 mol%) in toluene at −35°C) gave hydroxyester **3b** in 97% yield and 95% *ee* (Table 3, entry 5).

From previous reports that involve the most enantioselective ligands,[7b,c] the maintenance of high enantiocontrol across a variety of ketoesters still represents a challenge. For this reason, once the optimal reaction conditions were established, we wondered if this methodology was amenable to structural modification of the α-ketoester. To this end, asymmetric methylation of a series of substrates with different steric and electronic properties was studied. The results are summarized in Table 4.

It is highly remarkable that the efficiency of the enantiocontrol of our catalytic system was maintained at very high levels, independent of the nature of the R and R′ groups of the α-ketoester (Table 4, entries 1–13). As shown in Table 4, the enantiocontrol was not appreciably influenced by electronic or steric effects and the methylation of diverse *para*- and *ortho*-substituted aromatic, heteroaromatic, and aliphatic α-ketoesters was promoted with very good to excellent enantioselectivity. In this context, it was found that alkoxy substituents of different sizes, such as methoxy-, ethoxy-, and isopropoxy groups (**2a–c**, entries 1–3), were perfectly tolerated by our ligand and afforded the corresponding products with homogeneous efficiencies (94–97% yield) and enantioselectivities (93–95% *ee*). The introduction of electron-donating or electron-withdrawing substituents at the *para* position of the aryl group (for example, *p*-methyl, *p*-methoxy, and *p*-chloro (**2f–h**); Table 4, entries 6–8), did not affect the enantiocontrol of the addition and the respective products were isolated with high yields and similar *ee* values (91–94%).

To our delight, steric effects did not play an important role in the enantioselectivity either (Table 4, entries 4, 5, and 10). In contrast to the only previous report,[7e] *ortho*-substituted aromatic ketoesters **2e** and **2d** (Table 4, entries 4 and 5) underwent the asymmetric methyl addition with good to excellent enantioselectivities (96 and 85% *ee*, respectively), similarly to their *para* analogues. Even the bulky 2-naphthyl ketoester **2j** furnished the desired product **3j** with 93% yield and 91% *ee* (Table 4, entry 10). Notably, the use of *ortho*-substituted or electron-rich aromatic rings required longer reaction times to obtain moderate to good yields (Table 4, entries 4, 5, 7, and 9), although the unreacted ketoesters **2d**, **2e**, and **2i** were recovered unchanged.

The scope of the reaction was further explored and heteroaromatic (**2k**) and aliphatic (**2l**) ketoesters (Table 4, entries 11 and 12) and the cyclic furandione **2m** (Table 4, entry 13) were subjected to the methylation protocol. Products **3k–m**, respectively, were delivered with good enantioselectivity (84–90% *ee*) and high yields. Among the most challenging substrates reported to date in methyl transfer are aliphatic α-ketoesters, which are known to proceed with poor enantiocontrol. Shibasaki[7b] reported a complex mixture of products when the ketoester **2l** was used; Pedro[7d] and Uang[7e] reported moderate yields but low asymmetric induction for the same compound (35 and 19% *ee*, respectively). Hoveyda[7c] did not employ the same starting substrate, so comparison was not possible, although good enantioselectivity was achieved for a different aliphatic α-ketoester. As can be seen in Table 4, entry 12, the aliphatic ke-

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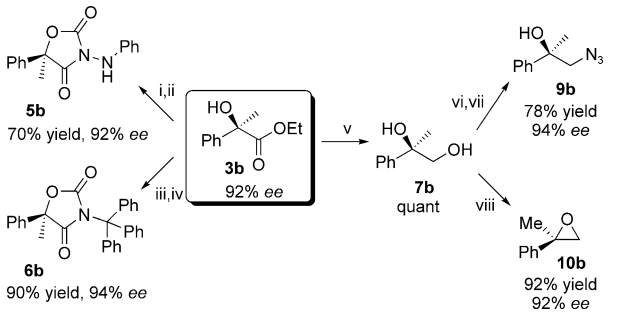
toester **2l** afforded the corresponding product **3l** with good enantiocontrol and excellent isolated yield (96% yield, 84% *ee*). Even the cyclic ketoester **2m** provided positive results when it was subjected to the methylation conditions (75% yield, 86% *ee*; Table 4, entry 13). Interestingly, the enantioselectivity was reversed for this substrate and the *S* enantiomer was obtained as the major product.

Ligand **1c** was recovered in over 90% yield after chromatographic purification. This recovered ligand was reused in the methylation of **2b** and produced essentially the same results as the freshly prepared ligand without any obvious decrease in chemical yield or enantiopurity (95% yield, 94% *ee*). In addition, the recovered ligand could be used to perform the same reaction on a 5 mmol scale and the desired product was isolated in 90% yield with 92% *ee*.

To demonstrate the potential value of the enantioenriched hydroxyesters obtained by methylation, we decided to carry out the synthesis of more complex and valuable structures, in which the quaternary stereocenter previously generated remained intact (Scheme 1). For example, azidoalcohols are




Scheme 1. Enantioselective synthesis of valuable building blocks from α-hydroxyester **3b**. Reagents and conditions: i) $ClCO_2Ph$, Py, $CH_2Cl_2$, RT overnight. ii) $PhNHNH_2$, DMAP, HAcO, THF, reflux, 36 h. iii) NaOEt, Urea, EtOH, reflux, 18 h. iv) TrCl, $NEt_3$, $CH_2Cl_2$, RT, 1 h. v) LAH, Et2O, 0°C, 30 min. vi) TsCl, $NEt_3$, DMAP, $CH_2Cl_2$, 0°C, 16 h. vii) $NaN_3$, $n$Bu$_4$NI, DMF, 80°C, 5 h. viii) TsCl, $NEt_3$, DMAP, $CH_2Cl_2$, 50°C, 24h.

valuable building blocks for the synthesis of aziridines and primary aminoalcohols.[9c] Conventional azidolysis of 2,2-disubstituted epoxides was found to provide regioisomeric mixtures of azidoalcohols, whereas enzymatic resolution of the racemic epoxides afforded the products with high *ee* values, although with poor chemical yields.[9c] Conversely, when our catalytic system was used the high-value azidoalcohol
(*R*)-**9b** was easily isolated in 76% overall yield with 94% *ee* (over four steps from commercially available **2b**). Additionally, the preparation of 2,2-disubstituted epoxide **10b** was accomplished in two simple steps in good yield. Two oxazolidinediones, **5b** and **6b**, were also prepared in two steps from the same hydroxyester **3b**,[9a,b] in 70 and 90% overall yield, respectively. It was confirmed by chiral HPLC that the quaternary stereocenter was not modified to any significant extent under the simple transformations described.

## Conclusion

We have shown that perhydro-1,3-benzoxazine-based ligand **1c** is an excellent promoter for the construction of quaternary stereocenters by enantioselective methylation of α-ketoesters by using $Me_2Zn$ as the alkylating agent and $B(OEt)_3$ as an additive. The enantiocontrol achieved with this catalytic system is high, and noticeably homogeneous, for a wide range of aromatic and heteroaromatic α-ketoesters, independent of the electronic nature of the substrate. The steric hindrance of *ortho*-substituted aromatic ketoesters is overcome by this ligand to afford the respective tertiary alcohols with up to 96% *ee*. Even the challenging cyclic and acyclic aliphatic α-ketoester substrates provide the desired hydroxyesters with good enantiocontrol. In addition, some interesting applications of the obtained hydroxyesters in the synthesis of valuable products are illustrated.

## Experimental Section

All reactions were carried out in anhydrous solvents under argon atmosphere in dried glassware by using Schlenk techniques. $^1$H (300 MHz) and $^{13}$C NMR (75 MHz) spectra were recorded in $CDCl_3$. Chemical shifts (δ) for are reported in ppm relative to tetramethylsilane with the residual solvent signal as an internal reference. Three $^{13}$C NMR spectra are shown for each compound (from top to bottom): DEPT90 (CH peaks only), DEPT135 (CH and $CH_3$ peaks up, $CH_2$ peaks inverted), and {$^1$H}-$^{13}$C. Data are reported as follows: chemical shift (δ, ppm), multiplicity (s = singlet, d = doublet, t = triplet, q = quartet, quint = quintet, sext = sextet sp = septet, m = multiplet, br = broad), coupling constants (*J*, Hertz) and integration. Specific rotations were measured with a 5 mL cell, 1 dm path length, and sodium lamp, and the concentration is given in g/100 mL. Flash chromatography was carried out on silica gel (230–240 mesh). Chemical yields refer to pure isolated substances. TLC analysis was performed on glass-backed plates coated with silica gel 60 and an $F_{254}$ indicator and plates were visualized either by UV irradiation or staining with $I_2$ or phosphomolybdic acid solution. Chiral HPLC analysis was performed by using a Daicel Chiralcel OD, Chiralpak AD-H, or Chiralpak AS-H column. UV detection was monitored at $\lambda = 220$ or 254 nm. HRMS was performed on a quadrupole spectrometer with TOF analyzer. Ketoesters **2a**, **2b**, **2l**, and **2m** were purchased from commercial sources and used as received. Ketoesters **2c** and **2d–k** were synthesized from diethyl oxalate and freshly prepared Grignard reagents by reported procedures.[7d,13] Racemic hydroxyesters were isolated by addition of methylmagnesium iodide (1.0 M in $Et_2O$, 1 equiv) to the corresponding ketoesters (1 equiv) in $Et_2O$ at −40°C. Ligands **1a**, **1b**, and **1d–f** were prepared by literature procedures.[8a,10] The reported synthesis of ligand **1c** was optimized.[10a]

**Typical procedure for enantioselective methylation of ketoesters**: A solution of $Me_2Zn$ (1.2 M, 2.5 mL, 3 mmol) in toluene was added dropwise to a solution of ligand **1c** (32.1 mg, 0.1 mmol) in anhydrous toluene (1.5 mL) under argon atmosphere at RT. The mixture was stirred for 10 min at RT, then triethyl borate (21 μL, 0.125 mmol) was added. The resulting mixture was stirred for another 5 min at RT, then was cooled to −35°C. At −35°C, the corresponding ketoester **2** (0.5 mmol) was added slowly (30 min) and the reaction mixture was stirred at −35°C for the indicated time. Afterwards, the mixture was quenched with saturated aqueous solution of $NH_4Cl$, extracted with $Et_2O$ (5 × 15 mL), dried over anhydrous $MgSO_4$, filtered, and the solvents were evaporated under reduced pressure. Purification by silica gel column chromatography (ethyl acetate/hexane) gave the pure hydroxyesters. Enantiomeric excess was determined by chiral HPLC.

**Compound 3a**: Compound **3a** was obtained from **2a** (73 μL, 0.5 mmol) after purification by flash chromatography (ethyl acetate/hexane 1:30) as

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a colorless liquid (94%). $[\alpha]_D^{20}=-48.1$ ($c=0.56$, $CHCl_3$) lit.[7c] $[\alpha]_D^{20}=-47.8$, $c=0.780$, $CHCl_3$, 91% for *R*; $^1$H NMR (300 MHz, $CDCl_3$): $\delta=1.79$ (s, 3H), 3.76 (s, 3H), 3.90 (brs, 1H; OH), 7.27–7.31( m, 1H), 7.35–7.38 (m, 2H), 7.55 ppm (d, $J=7.2$ Hz, 2H); $^{13}$C NMR (75 MHz, $CDCl_3$): $\delta=26.6$ ($CH_3$), 53.1 ($CH_3$), 75.7 (C), 125.0 (2×CH), 127.7 (CH), 128.2 (2CH), 142.6 (C), 176.0 ppm (C); IR (neat): $\tilde{\nu}=3493$, 1726, 1250, 1144, 1096, 1070, 732, 697, 658 cm$^{-1}$; MS (EI): *m/z* (%): 180 [$M^+$] (30), 151 (62), 121 (100), 105 (45), 77 (28); HRMS calcd for $C_{10}H_{12}O_3Na$: 203.0679; found: 203.0677; HPLC (Chiralpak AD-H, hexane/isopropanol=98:2, 1 mL min$^{-1}$, $\lambda=220$ nm): retention time ($t_R$)=13.8 min for the *S* enantiomer, $t_R=17.0$ min for the *R* enantiomer (configuration was assigned by comparison of the sign of optical rotation and HPLC elution order[7b] with the literature data).

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